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VEHICLE BODY PANELS FOR SPEAKERS, SPEAKER ASSEMBLIES WITH VEHICLE BODY PANELS, AND INSTALLATION METHODS THEREFORE

Abstract

A vehicle body panel is provided defining an internal cavity that is acoustically sealed at least partially along a perimeter with an opening in the vehicle body panel for access to the internal cavity and sized to receive a speaker in the opening. A speaker assembly is provided with the speaker is installed in the opening. A speaker assembly is provided with a first vehicle body panel with an opening. A speaker is installed in the opening. A second vehicle body panel is attached to the first vehicle body panel and sealed along a connection with the first vehicle body panel to collectively provide an internal cavity accessible at the opening, to provide a resonation chamber for the speaker. A method is provided by sealing a plurality of vehicle body panels. A speaker is installed in an opening in the plurality of sealed vehicle body panels.

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Background/Summary

TECHNICAL FIELD

[0001] Various embodiments relate to vehicle body panels, speaker assemblies, and installation methods, therefore.

BACKGROUND

[0002] The prior has provided speaker assemblies in vehicle bodies by installing speaker assemblies to trim components that are attached to the vehicle body.

SUMMARY

[0003] According to an embodiment, an apparatus is provided as a vehicle body panel defining an internal cavity that is acoustically sealed at least partially along a perimeter with an opening in the vehicle body panel for access to the internal cavity and sized to receive a speaker in the opening.

[0004] According to a further embodiment, the internal cavity provides a resonance chamber for the speaker.

[0005] According to an even further embodiment, a port is formed through the vehicle body panel to port the resonance chamber.

[0006] According to an even further embodiment, the port further provides a tube.

[0007] According to another even further embodiment, the port is vented externally of the vehicle body panel.

[0008] According to another further embodiment, the vehicle body panel is formed from sheet metal.

[0009] According to another embodiment, a vehicle body panel is provided defining an internal cavity that is acoustically sealed at least partially along a perimeter with an opening in the vehicle body panel for access to the internal cavity and sized to receive a speaker in the opening. The vehicle body panel is further defined as a first vehicle body panel. A second vehicle body panel is attached to the first vehicle body panel to collectively provide the internal cavity.

[0010] According to a further embodiment, the second vehicle body panel is welded to the first vehicle body panel.

[0011] According to another further embodiment, the first vehicle body panel and the second vehicle body panel collectively provide a pillar.

[0012] According to another further embodiment, a third vehicle body panel is connected to the first vehicle body panel and the second vehicle body panel to enclose a region between the first vehicle body panel and the second body panel to at least partially enclose the internal cavity.

[0013] According to an even further embodiment, a port is formed through the third vehicle body panel to port the internal cavity.

[0014] According to another embodiment, a speaker assembly is provided with a vehicle body panel defining an internal cavity that is acoustically sealed at least partially along a perimeter with an opening in the vehicle body panel for access to the internal cavity and sized to receive a speaker in the opening. The speaker is installed in the opening.

[0015] According to a further embodiment, the speaker is further provided as a cone speaker.

[0016] According to another further embodiment, the speaker is further provided as a sub-woofer.

[0017] According to another further embodiment, the speaker extends at least partially into the internal cavity.

[0018] According to another further embodiment, a resonance chamber is provided, wherein the speaker is installed to an opening in the resonance chamber, and wherein the resonance chamber is installed in the opening in the vehicle body panel.

[0019] According to an even further embodiment, the resonance chamber extends at least partially into the internal cavity.

[0020] According to another embodiment, a speaker assembly is provided with a first vehicle body panel with an opening. A speaker is installed in the opening. A second vehicle body panel is attached to the first vehicle body panel and sealed along a connection with the first vehicle body panel to collectively provide an internal cavity accessible at the opening, to provide a resonation chamber for the speaker.

[0021] According to another embodiment, a method is provided by sealing a plurality of vehicle body panels. A speaker is installed in an opening in the plurality of sealed vehicle body panels.

[0022] According to a further embodiment, a port is formed in the plurality of sealed vehicle body panels.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0023] FIG. 1 is a top exterior view of a partially assembled vehicle body according to an embodiment;

[0024] FIG. 2 is a front perspective view of an interior of the partially assembled vehicle body of FIG. 1 facing a rear region of the vehicle body;

[0025] FIG. 3 is an enlarged partial front section view of a first side of the partially assembled vehicle body of FIG. 1;

[0026] FIG. 4 is an enlarged partial side perspective view of the interior of a second side of the partially assembled vehicle body of FIG. 1;

[0027] FIG. 5 is an enlarged partial side perspective view of the exterior of the partially assembled vehicle body of FIG. 1;

[0028] FIG. 6 is an enlarged front partial section view of the second side of the partially assembled vehicle body of FIG. 1;

[0029] FIG. 7 is an enlarged rear partial section view of the second side of the partially assembled vehicle body of FIG. 1;

[0030] FIG. 8 is an enlarged front partial section view of the second side of the partially assembled vehicle body of FIG. 1;

[0031] FIG. 9 is an enlarged exterior perspective view of the second side of the partially assembled vehicle body of FIG. 1;

[0032] FIG. 10 is another enlarged exterior perspective view of the second side of the partially assembled vehicle body of FIG. 1;

[0033] FIG. 11 is an enlarged front partial second view of the second side of the partially assembled vehicle body of FIG. 1;

[0034] FIG. 12 is a side perspective exterior view of a rear region of a partially assembled vehicle body according to another embodiment;

[0035] FIG. 13 is an enlarged side perspective exterior view of the rear region of the partially assembled vehicle body of FIG. 12; and

[0036] FIG. 14 is an enlarged side perspective interior view of the rear region of the partially assembled vehicle body of FIG. 12.

DETAILED DESCRIPTION

[0037] As required, detailed embodiments of the present invention are disclosed herein; however, it is to be understood that the disclosed embodiments are merely exemplary of the invention that may be embodied in various and alternative forms. The figures are not necessarily to scale; some features may be exaggerated or minimized to show details of particular components. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a representative basis for teaching one skilled in the art to variously employ the present invention.

[0038] FIG. 1 is a top view of a partially assembled vehicle body 30. The vehicle body 30 is illustrated as one example of a vehicle body 30. For example, the vehicle body 30 is the body 30 of a passenger land vehicle. The vehicle body 30 may be any vehicle, such as any land vehicle, watercraft, aircraft, or the like. The vehicle body 30 may also be a straddle type vehicle, cargo vehicle, passenger vehicle, or the like.

[0039] The vehicle body 30 is illustrated as sheet metal, often referred to as body-in-white. The vehicle body 30 may be formed from other materials, such as polymeric materials, carbon fiber reinforced materials, or the like. The vehicle body 30 may also be formed from a plurality of vehicle body panels as a vehicle body panel assembly 30.

[0040] In the depicted embodiment, directions are referred to from the perspective of an external viewer, and not from the perspective of an internal passenger. For example, a direction of a front view 32 is depicted by arrow 32. In contrast, the front view 32 is opposite to a direction of travel of the vehicle. The description of directions in FIG. 1 are aligned with the external viewer as discussed by an installer, not an internal occupant within the vehicle body 30. Therefore, the front view 32 is in a direction from a front of the vehicle to a rear of the vehicle.

[0041] The vehicle body 30 includes a first side 34 and a second side 36. The sides 34, 36 are labeled by order of introduction in the figures and do not denote any other difference. For example, the first side 34 is illustrated as a right side, when described from the exterior, which may be a driver side in North America, or a passenger side in other jurisdictions. The second side is illustrated as a left side, when described from the exterior, which may be a passenger side in North America. Each vehicle body side 34, 36 may be formed from stamped sheet metal as is known in the art for attachment to a vehicle chassis. The vehicle body sides 34 and 36 are generally spaced apart to enclose and define a vehicle interior 38. The vehicle interior 38 includes a passenger compartment 40 and a cargo compartment 42. The compartments 40, 42 are by way of illustration and example. In other embodiments, the vehicle interior 38 may include only a passenger compartment 40, or a cargo compartment 42. The vehicle interior 38 may receive vehicle interior components therein, such as passenger seating, passenger restraints and safety components that are not illustrated, for example, seat assemblies, seatbelts, airbags, and the like.

[0042] The vehicle sides 34, 36 each include a lower rail 44, 46 and an upper rail 48, 50 interconnected by a plurality of pillars 52, 54, 56, 58, 60, 62, 64, 66. The first pillars 52, 54 may be formed integrally with the upper rails 48, 50, and may be referred to as A-pillars 52, 54. The first pillars 52, 54 are spaced apart from the second pillars 56, 58 to provide door openings for occupant ingress into, and egress from, the passenger compartment 40 of the vehicle interior 38. The second pillars 56, 58 may be referred to as B-pillars 56, 58. The third pillars 60, 62 are spaced apart from the second pillars 56, 58 to provide door openings for occupant ingress into, and egress from, the passenger compartment 40 the vehicle interior 38. The third pillars 60, 62 may be referred to as C-pillars 60, 62. The fourth pillars 64, 66 are spaced apart from the third pillars 60, 62 to provide a window opening within the cargo compartment 42. The fourth pillars 64, 66 may be referred to as D-pillars 64, 66. The fourth pillars 64, 66 may converge with the third pillars 60, 62 at the roof rails 48, 50.

[0043] Conventionally, trim components are installed upon the pillars 52, 54, 56, 58, 60, 62, 64, 66 within the vehicle interior 38. Various comfort and entertainment features are installed into the trim components, including speaker assemblies. Speaker assemblies are often provided with a speaker cone attached to a speaker box, or a resonation chamber, which is an enclosed or partially enclosed chamber to provide a resonating back pressure to the speaker for quality of acoustics and sound output. Resonation chambers for larger speakers, such as subwoofers, often include a port for intake and exhaust of back pressure to a rear side of the speaker.

[0044] Speaker assemblies are often installed in the trim components in the vehicle interior. Vehicle manufacturers, or original equipment manufacturers (OEMs), have been reducing an overall size of vehicles in order to reduce energy requirements of vehicles, and increase efficiencies of vehicles,

such as fuel efficiency. In order to maximize and/or optimize the vehicle interior **38**, OEMs have been increasing a width of the vehicle interior **38**. The increased width of the vehicle interior **38** moves the trim members closer to the pillars **52, 54, 56, 58, 60, 62, 64, 66**.

[0045] In order to increase the width of the vehicle interior **38**, or maintain the width of the vehicle interior **38**, a speaker assembly **68** is provided that provides space optimization in comparison to the prior art and is illustrated in FIGS. **3-11**. The speaker assembly **68** is integrated into the vehicle body **30** and is not housed solely within a trim member (not shown). As depicted, the speaker assembly **68** includes a cone speaker **70** installed directly to the third pillar **60**. In the depicted embodiments, the third pillar **60** is formed from at least panels **72, 74** to define a cavity **76** within the panels **72, 74**. Although the third pillar **60** is depicted as two separate panels **72, 74**, the third pillar **60** may be formed integrally, such as a hydroformed column, or the like. Although the speaker assembly **68** is illustrated and described in conjunction with the third pillars **60, 62**, the speaker assembly **68** may be incorporated and installed into any panel of the vehicle body **30**.

[0046] An opening **78** is formed into the interior panel **72** that is sized to receive the speaker **70** as illustrated in FIGS. **6-11** to provide access to the cavity **76**. The speaker **70** includes a flange **80** that is fastened to the interior panel **72** to mount the speaker **70**, and to seal the opening **78** as illustrated in FIGS. **4, and 6-11**. The speaker **70** extends at least partially into the cavity **76**. The cavity **76** is sealed at least partially along a perimeter of the cavity **76** to control a fluid flow of air pressure within the cavity **76**. According to an embodiment, the third pillars **60, 62** are formed from sheet metal that is sealed about the cavity **76**, such as by a weld seam, a sealant, or the like. The sealed enclosure of the cavity **76** provides a resonation chamber **76** for the speaker assembly **68**.

[0047] By utilizing panels **72, 74** of the third pillars **60, 62** as the resonation chamber **76**, a separate resonation chamber **76** is not required, that conventionally was housed within trim components within the vehicle interior **38**. Therefore, the speaker assembly **68** reduces components by utilizing existing components of the vehicle body **30** while optimizing space within the vehicle interior **38**. Referring now to FIG. **3**, the resonation chamber **76** is no longer housed within a dimension A within trim components. Instead, the resonation chamber **76** is formed within a dimension B, thereby permitting the trim components to be smaller for a reduction of dimension A.

[0048] FIG. **5** illustrates the third pillar **62** from the exterior of the vehicle body **30**. The pillar **62** may include a plurality of ports **82, 84**, which may be provided for welding clearance or ventilation, external access to the cavity **76**, or any other manufacturing consideration. According to an embodiment, the ports **82, 84** are sealed with plugs or caps to enclose the cavity **76**.

[0049] Referring now to FIGS. **1-11**, the third pillars **60, 62** extend from the roof rails **48, 50** to wheel wells **86, 88**, which provide clearance for rear wheels of the vehicle. For particular speakers **70**, such as subwoofers **70** or external couple subwoofers **70**, a port **90** may be formed through the vehicle body **30** to exhaust the chamber **76** to the exterior of the vehicle body **30**. The port **90** is illustrated in FIGS. **6-9** and **11**, formed through the wheel wells **86, 88**. The port **90** permits ventilation for the resonation chamber **76** for tuning the resonation chamber **76** to balance a back pressure for the speaker **70** by exhaust of pressure in a direction D illustrated in FIG. **7**.

[0050] Vehicle pillars **52, 54, 56, 58, 60, 62, 64, 66** vary in size, and therefore vary in the size of the cavity **76** therein. Speakers **70** also vary in the resonation chamber **76** size requirements. Referring to FIG. **7**, an additional insert panel **92** may be provided in the cavity **76** to reduce the volume of the resonation chamber **76** to a region labeled C. The insert panel **92** extends from the exterior panel **74**, toward the speaker **70** and then above the opening **78**, and then to the interior panel **72**. The insert panel **92** may be welded or otherwise sealed to the interior and exterior panels **72, 74**.

[0051] FIGS. **8** and **9** illustrate an insert panel **94** according to another embodiment. The insert panel **94** is formed from a flat sheet of material, such as sheet metal. The insert panel **94** is oriented above the speaker **70**, and is welded or otherwise sealed within the cavity **76** to the pillar panels **72, 74**. Insert panels **92, 94** of various sizes, shapes, and installation locations permit sizing of the

resonation chamber **76** tuned to the particular speaker **70** selected for the application.

[0052] FIGS. **10** and **11** illustrate that the port **90** may include a tube **96**. The tube **96** is welded into the port **90** to extend into the resonation chamber **76** for speaker assemblies **68** that specify the tube **96** within the chamber **76** for tuning the back pressure.

[0053] FIGS. **12-14** illustrate the vehicle body **30** for application of a speaker assembly **102** according to another embodiment. The vehicle body **30** includes an interior rear quarter panel **104**, **106** for each vehicle body side **34**, **36**. Each rear quarter panel **104**, **106** is bounded by the corresponding third and fourth pillars **60**, **62**, **64**, **66** and the corresponding wheel well **86**, **88**. FIG. **12** illustrates the vehicle body **30** with a decorative skin or decorative panel uninstalled. The vehicle body **30** may include an outboard quarter panel frame **108**. The interior rear quarter panel **106** and the outboard quarter panel frame **108** collectively provide a cavity **110** therebetween.

[0054] FIGS. **13** and **14** illustrate the speaker assembly **102** partially oriented within the cavity **110** for each side **34**, **36** of the vehicle body **30**. An opening **112**, **114** is formed through each interior rear quarter panel **104**, **106**. The openings **112**, **114** permit access to the cavity **110** of each vehicle body side **34**, **36**. The speaker assembly **102** of each side includes a resonation chamber **116**, **118** that is oriented primarily within the cavity **110**, and at least partially through the respective opening **112**, **114**. The resonation chamber **116**, **118** provides a speaker box or enclosure, and may be formed from a polymeric material, or any suitable structural material. The resonation chambers **116**, **118** may be installed in the vehicle during, or after the assembly of the vehicle body **30**.

[0055] The speaker assembly **102** includes a cone speaker **120** installed to an opening **122** in the resonation chamber **116**, **118** to face into the vehicle interior **38**. The speaker assembly **102** orients the majority of the resonation chamber **116**, **118** and the speaker **120** into the interior rear quarter panel **104**, **106** to minimize a thickness of a trim cover upon the interior rear quarter panel **104**, **106** and to maximize the size of the vehicle interior **38**. The speaker assembly **102** may be preassembled prior to assembly to the vehicle body **30**.

[0056] While various embodiments are described above, it is not intended that these embodiments describe all possible forms of the invention. Rather, the words used in the specification are words of description rather than limitation, and it is understood that various changes may be made without departing from the spirit and scope of the invention. Additionally, the features of various implementing embodiments may be combined to form further embodiments of the invention.

Claims

1. An apparatus comprising a vehicle body panel defining an internal cavity that is acoustically sealed at least partially along a perimeter with an opening in the vehicle body panel for access to the internal cavity and sized to receive a speaker in the opening.
2. The apparatus of claim 1 wherein the internal cavity provides a resonation chamber for the speaker.
3. The apparatus of claim 2 further comprising a port formed through the vehicle body panel to port the resonation chamber.
4. The apparatus of claim 3 wherein the port further comprises a tube.
5. The apparatus of claim 3 wherein the port is vented externally of the vehicle body panel.
6. The apparatus of claim 1 wherein the vehicle body panel is formed from sheet metal.
7. A vehicle body panel assembly comprising: the apparatus of claim 1, wherein the vehicle body panel is further defined as a first vehicle body panel; and a second vehicle body panel attached to the first vehicle body panel to collectively provide the internal cavity.
8. The vehicle body panel assembly of claim 7 wherein the second vehicle body panel is welded to the first vehicle body panel.
9. The vehicle body panel assembly of claim 7 wherein the first vehicle body panel and the second vehicle body panel collectively provide a pillar.

- 10.** The vehicle body panel assembly of claim 7 further comprising a third vehicle body panel connected to the first vehicle body panel and the second vehicle body panel to enclose a region between the first vehicle body panel and the second vehicle body panel to at least partially enclose the internal cavity.
 - 11.** The vehicle body panel assembly of claim 10 further comprising a port formed through the third vehicle body panel to port the internal cavity.
 - 12.** A speaker assembly comprising: the apparatus of claim 1; and the speaker installed in the opening.
 - 13.** The speaker assembly of claim 12 wherein the speaker further comprises a cone speaker.
 - 14.** The speaker assembly of claim 12 wherein the speaker further comprises a sub-woofer.
 - 15.** The speaker assembly of claim 12 wherein the speaker extends at least partially into the internal cavity.
 - 16.** The speaker assembly of claim 12 further comprising a resonation chamber, wherein the speaker is installed to an opening in the resonation chamber, and wherein the resonation chamber is installed in the opening in the vehicle body panel.
 - 17.** The speaker assembly of claim 16 wherein the resonation chamber extends at least partially into the internal cavity.
 - 18.** A speaker assembly comprising: a first vehicle body panel with an opening; a speaker installed in the opening; and a second vehicle body panel attached to the first vehicle body panel and sealed along a connection with the first vehicle body panel to collectively provide an internal cavity accessible at the opening, to provide a resonation chamber for the speaker.
 - 19.** A method comprising: sealing a plurality of vehicle body panels; and installing a speaker in an opening in the plurality of sealed vehicle body panels.
 - 20.** The method of claim 19 further comprising: forming a port in the plurality of sealed vehicle body panels.
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